

SPRING BOOT

LOOSE COUPLING EXAMPLE:

LOOSE COUPLING APPLICATION

```
package com.example.LooseCoupling;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.ApplicationContext;

@SpringBootApplication
public class LooseCouplingApplication {

    public static void main(String[] args) {
        ApplicationContext context= SpringApplication.run(LooseCouplingApplication.class, args);
        Student s = context.getBean(Student.class);
        s.writeExam();
    }

}
```

STUDENT

```
package com.example.LooseCoupling;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.beans.factory.annotation.Qualifier;
import org.springframework.stereotype.Component;

@Component 2 usages
public class Student {
    @Autowired
    public Student(@Qualifier("sketch") WriterTool tool){
        this.tool = tool;
    }
    private WriterTool tool; 2 usages
    public void writeExam(){ 1 usage
        System.out.println("Student is writing");
        tool.write();
    }

}
```

PEN

```
package com.example.LooseCoupling;

import org.springframework.context.annotation.Primary;
import org.springframework.stereotype.Component;

@Component
@Primary
public class Pen implements WriterTool {
    public void write(){ 1 usage
        System.out.println("Writing with Pen");
    }
}
```

PENCIL

```
package com.example.LooseCoupling;

import org.springframework.stereotype.Component;

@Component
public class Pencil implements WriterTool {
    public void write() 1 usage
        System.out.println("Writing with Pen");
    }
}
```

SKETCH

```
package com.example.LooseCoupling;

import org.springframework.stereotype.Component;

@Component
public class Sketch implements WriterTool {
    public void write(){ 1 usage
        System.out.println("Writing with Sketch");
    }
}
```

WRITERTOOL

```

package com.example.LooseCoupling;

import org.springframework.stereotype.Component;

@Component
public class Sketch implements WriterTool {
    public void write(){ 1 usage
        System.out.println("Writing with Sketch");
    }
}

```

OUTPUT:

```

2026-08-27T09:46:19.674+05:30 INFO 37428 --- [Loose
Student is writing
Writing with Sketch

```

WITHOUT QUALIFIER:

```

package com.example.LooseCoupling;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.beans.factory.annotation.Qualifier;
import org.springframework.stereotype.Component;

@Component 2 usages
public class Student {
    @Autowired
    public Student( WriterTool tool){
        this.tool = tool;
    }

    private WriterTool tool; 2 usages
    public void writeExam(){ 1 usage
        System.out.println("Student is writing");
        tool.write();
    }
}

```

OUTPUT:

```

2026-08-27T09:45:16.452+05:30 INFO 37084 ---
Student is writing
Writing with Pen

```